

AFTER THE COLLAPSE

Northern cod · NAFO 2J3KL

the 1992 moratorium
can any catch policy hold the LRP?

the fitted surplus map
 $r = 0.2369 \cdot K = 5000 \text{ kt}$ (at bound)

limit reference point
884.6 kt (2016) — a floor on SSB

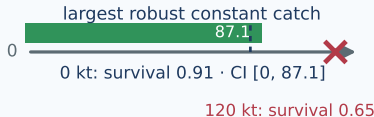
protocol frozen before any score
kernels · boundaries · replays · selection

WHAT CATCH CAN HOLD

under the worst-case floor

worst-case productivity floor
the 10th-percentile class

91.6 kt



zero catch and the moratorium
hold the safe set; the critical-zone and cascade rules hold the LRP from itself

WHEN CATCH CANNOT HELP

the map is expansive at the LRP

no non-BAU policy dominates BAU
under the declared partial order

positive-catch rules: empty
at the 5th-percentile floor, $T = \infty$

certified kernels: empty
beyond 7 yr · $K \geq 2K^* = 1769.2 \text{ kt}$

the margin good years must supply is
smaller than the frozen convention implied
the LRP is protected by good years

Robust Viability of the 2J3KL Limit Reference Point under a Surplus-Production Map: Policy Scoring, Expansion, and When Catch Cannot Help